Current makes a magnetic field

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect current direction with the magnetic field around a wire.

SUCCESS CRITERIA - I can...

1. Sketch circular field lines around a straight wire.
2. Use a stated right-hand rule.
3. Predict the effect of reversing current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “current” and “magnetic field” to explain your answer. Sentence starters are optional.

current: rate of charge flow

magnetic field: a field that influences magnets and moving charges

radius: distance from the center to the edge of a circle or sphere

01 Warm-up: bring back an earlier idea

A total current of **12 A** splits between two identical parallel branches. **Find each branch current**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $B = \mu_0 I/(2\pi r)$; $\mu_0 = 4\pi \times 10^{-7}$ **T·m/A**.
Long straight wire. Right thumb follows conventional current; curled fingers show field direction.

Worked reasoning: Around a straight wire carrying conventional current out of the page, the magnetic field circles counterclockwise by the right-hand rule. A dot means out of page.

Your turn: Draw three field arrows around the wire and **explain what the dot represents**.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .

Current out of page (dot). Add field arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Conventional current points out of the page. Use the right-hand rule to **draw the surrounding magnetic field** direction.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Current out of page (dot). Add field arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

A long straight wire carries current out of the page. At a fixed distance, the current doubles.
1. Use $B \propto I$ to **compare field strengths**.
2. **Add field arrows** and indicate whether their direction changes.

Sentence starter: Both ... . They differ in ... because ... .

Current out of page (dot). Add field arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

What happens to field direction if the current reverses while its magnitude stays the same?

Word help: magnitude = size of a quantity without its direction

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two objects have charges **-4 nC** and **-6 nC**. **Predict attraction or repulsion** and **draw both force directions**.
Label charges and both interaction forces.

Sentence starter: I predict … . I used … to make this prediction because … .



Charge A Charge B

Marzano check - circle: 1 2 3

07 Exit check

At the same current, distance from a long wire doubles. Using B proportional to **1**/r, compare field strength.

Sentence starter: Both … . They differ in … because … .

Current out of page (dot). Add field arrows.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**. Reverse the current. Redraw the field arrows and **explain what stays the same** in an ideal symmetric setup.

Word help: ideal = using stated simplifying conditions

Sentence starter: I think … . I can point to …, which shows …, so … .

Cross = current into page. Add magnetic-field arrows.



Marzano check - circle: 1 2 3